

Wuhan University 2022–2023 Spring Introduction to Data Science Final Exam

(Total: 100 points Exam duration: 150 minutes)

1. (20 points)

Suppose we have data $x_1, \dots, x_n$ with each $x_i \in \mathbb{R}^d$ and d is huge. We want to compress the data by linear map: i.e., find linear encoder $E \in \mathbb{R}^{s \times d}$ and decoder $D \in \mathbb{R}^{d \times s}$ with $s \ll d$ that minimize the reconstruction loss

$$\mathcal{L}(E, D) = \frac{1}{n} \sum_{i=1}^n \|x_i - DEx_i\|_2^2.$$

What is the optimal E and D ?

2. (20 points)

Show that:

1. $x^* \in \arg \min_{x \in \mathbb{R}^n} f(x)$ if and only if $0 \in \partial f(x^*)$.
2. Let $\hat{x} = \arg \min_{x \in \mathbb{R}} f(x) = \frac{1}{2}(x - t)^2 + \lambda|x|$ with $t \in \mathbb{R}, \lambda \in \mathbb{R}_{++}$. Then

$$\hat{x} = S_\lambda(t) = \text{sign}(t) \max(|t| - \lambda, 0).$$

3. $\hat{x} = \arg \min_{x \in \mathbb{R}^n} f(x) = \frac{1}{2}\|Ax - b\|_2^2 + \lambda\|x\|_1$ with $b \in \mathbb{R}^n, \lambda \in \mathbb{R}_{++}$ if and only if

$$\hat{x} = S_{\lambda\tau}(\hat{x} + \tau A^\top(b - A\hat{x})), \quad \forall \tau > 0.$$

3. (20 points)

Prove the **Bounded Difference Inequality**:

Let $X_1, \dots, X_n$ be independent random variables. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a measurable function. Assume that $|f(x) - f(x')| \leq c_i$, where x' is an arbitrary change of the i -th coordinate of x , for $i = 1, \dots, n$. Then, for any $t > 0$, we have

$$\mathbb{P}[f(X) - \mathbb{E}f(X) \geq t] \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right),$$

where $X = (X_1, \dots, X_n)$.

4. (10 points)

Define the target function as

$$f^* \in \arg \min_{\text{measurable } f} \mathcal{R}(f) = \mathbb{E}_{(X,Y) \sim P} [\ell(f(X), Y)].$$

Show that:

1. $f^*(x) = \arg \min_{a \in \mathcal{Y}} \mathbb{E} [\ell(a, Y) \mid X = x]$.
2. For regression: $\mathcal{Y} = \mathbb{R}$, $\ell(a, y) = \frac{1}{2}(a - y)^2$, prove that $f^*(x) = \mathbb{E}[Y \mid X = x]$.
3. For classification: $\mathcal{Y} = [K] := \{1, 2, \dots, K\}$, $\ell(a, y) = \mathbf{1}_{a \neq y}$, prove that $f^*(x) = \arg \max_{k \in [K]} \mathbb{P}(Y = k \mid X = x)$.

5. (20 points)

Consider the linear model $Y = X\theta^* + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ and $n \ll d$. Define the Lasso estimator:

$$\hat{\theta}_L = \arg \min_{\theta \in \mathbb{R}^d} \left\{ \frac{1}{2n} \|Y - X\theta\|_2^2 + \lambda \|\theta\|_1 \right\}.$$

Let

$$\gamma = \inf_{v \in C} \frac{v^\top \frac{X^\top X}{n} v}{\|v\|_2^2} > 0,$$

with $C = \{v : \|v_{A^*}\|_1 \leq 3\|v_{A^*}\|_1\}$ and $A^* = \text{supp}(\theta^*)$. Set

$$\lambda = 2\sigma \sqrt{\frac{2 \ln(2d/\delta)}{n}}.$$

Prove that, with probability at least $1 - \delta$,

$$\|\hat{\theta}_L - \theta^*\|_2 \leq \frac{6}{\gamma} \sqrt{2 \ln(2d/\delta)} \sigma \sqrt{\frac{|A^*|}{n}}.$$

6. (10 points)

Write down your suggestion for this course.